

Compute estimate — prune-and-distill from an open teacher,
with prabhāsa’s Sanskrit reasoning folded in

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Status: decision support, not an ADR — no decision has been made to pursue this path.

Supersedes the “via teacher” analysis in `docs/qwen3-class-pretraining-estimate.md` §4, which modeled distillation as a from-scratch student trained alongside a teacher (no proven discount, per F53). This document instead prices NVIDIA’s own published **prune-and-distill** technique (Minitron), a fundamentally different and far cheaper regime, and adds a plan to fold **prabhāsa-samskrutam**’s Sanskrit/kāraka/Nyāya reasoning work in as a low-cost second stage rather than discarding it.

Author context: produced by an agent session in worktree `kinetic-ai-runpod-estimate-76abc3` on operator request. Not reviewed or acted on.

1 What changed since the last estimate

The prior document priced two paths: full from-scratch pretraining (six-to-seven figures) and teacher-guided from-scratch training (no proven discount per F53, likely *more* expensive than from-scratch). Neither gets near the \$20-30K target.

A third path exists and is NVIDIA’s own published production technique for exactly this class of model, not a speculative architecture discount: **structured pruning of an already-trained, already-competitive larger model, followed by continued training with knowledge distillation from the original (unpruned) model as teacher**. This is categorically different from training a student from scratch against a teacher — the student starts with the teacher’s weights (subsetting), not random initialization, so it needs vastly fewer tokens to recover quality.

2 Published numbers (not extrapolated)

- **Nemotron-Mini-4B-Instruct** (`nvidia/Nemotron-Mini-4B-Instruct` model card) is a fine-tune of `nvidia/Minitron-4B-Base`, itself pruned and distilled from Nemotron-4-15B. It was never pretrained from scratch.
- **Minitron** (arXiv 2408.11796, “LLM Pruning and Distillation in Practice”): pruning Llama-3.1-8B down to 4B and continuing training with distillation used **94 billion tokens** — a **150x reduction** against the 15T tokens the 8B teacher itself consumed. Reported result: up to 16% better MMLU than an equivalent-size model trained from scratch. NVIDIA’s summary states the *family-wide* compute saving (parent + all pruned children) is 1.8x; the *marginal* cost of each additional pruned-and-distilled child is the 40–150x-fewer-tokens figure.
- **Nemotron-H-8B** (arXiv 2504.03624), a from-scratch hybrid Mamba-Transformer, trained on **1 trillion tokens** — 125 tokens/param, notably below the ~1,875 tokens/param this project previously anchored to Llama-3-8B. Reported on-par-or-better accuracy vs Qwen-2.5-7B/72B and Llama-3.1-8B/70B.

- **No dollar training cost is published anywhere found for any of these models.** All \$ figures below are derived from the disclosed token counts using this project’s own FLOPs/GPU-cost methodology (§3), not sourced from NVIDIA.

Sources: <https://huggingface.co/nvidia/Nemotron-Mini-4B-Instruct> · <https://arxiv.org/abs/2408.11796> · <https://www.superannotate.com/blog/llm-pruning-distillation-minitron-approach> · <https://arxiv.org/abs/2504.03624>

3 Method

Same $C \approx 6ND$ base as prior estimates, 35% MFU assumption, same RunPod GPU catalog. Compute for the distillation stage is student-own-training plus teacher-inference overhead:

$$C \approx 6N_{\text{student}}D + 2N_{\text{teacher}}D$$

Token budget $D = 94\text{B}$, Minitron’s own disclosed figure for an 8B→4B pruning — used directly, not scaled, since it is the only concrete anchor available; §5 flags where this may not transfer. Teacher assumed at ~7–8B params (an existing open, ideally Apache-licensed, model — see §5).

4 Cost by target size and GPU

Target	Compute (FLOPs)	H100 Community	H100 Secure	RTX 5090 Comm. / Secure
1B	2.07e21	\$4,463	\$5,458	\$7,970 / \$11,435
1.7B (PRD target)	2.46e21	\$5,318	\$6,504	\$9,491 / \$13,617
3B	3.20e21	\$6,900	\$8,439	\$12,317 / \$17,671

All cells land under the \$20–30K target — the first estimate in this line of analysis that clears the bar, and it rests on a reproducible published technique rather than an assumed discount.

Folding in prabhasa-samskrutam’s Sanskrit reasoning work

Prabhasa’s own successful training stages ran on 350M–5.25B tokens per stage (M2: ~650M, M3: 350M–780M, M4: 5.25B). Adding a Sanskrit/kāraṇa/Nyāya continued-training stage on top of the pruned-and-distilled model, at a comparable 5B-token scale, for the 1.7B target:

$$C_{\text{sanskrit stage}} = 6 \times 1.7\text{e}9 \times 5\text{e}9 = 5.1\text{e}19 \text{ FLOPs}$$

roughly **2% of the main 2.46e21 distillation budget** — a few hundred dollars, not a material addition. This is the mechanism for keeping prabhasa’s Sanskrit-grounded reasoning as a genuine product differentiator rather than discarding it: the model gets pruned-and-distilled English competence from an existing capable teacher, then a cheap second stage on prabhasa’s kāraṇa-role/Nyāya-verification material layered on top.

5 What this estimate does not prove

1. **Compression ratio risk.** Minitron’s tested ratios were 15B→8B (1.9x) and 8B→4B (2x). This estimate applies the 94B-token budget to going as far as 8B→1B (8x) — well beyond anything published. Quality risk rises with compression ratio; the 3B target (8B→3B, 2.7x) is closer to tested territory and the more defensible of the three sizes. The 1B figure should be read as optimistic, not established.
2. **Teacher licensing** needs checking before committing: Qwen2.5-7B (Apache 2.0, clean), Llama-3.1-8B (Llama license, usage restrictions), Nemotron-H-8B (NVIDIA Open Model License).
3. **Engineering is real but not quantified here.** NVLabs’ Minitron pruning code is open-sourced (github.com/NVlabs/Minitron) and reusable rather than build-from-scratch, but adapting it to this project’s stack, plus integrating prabhasa’s Nyāya-verifier and Megatron-Core/Nemotron-H tooling, is unquantified integration time.
4. **No SFT/RLHF cost included** — pretraining-adjacent compute alone does not ship a benchmark-ready instruct model, consistent with every prior estimate in this line of work.
5. **94B tokens is one data point, not a scaling law.** It has not been shown to hold at different compression ratios or target sizes; treat it as the best available anchor, not a guarantee.
6. These figures are extrapolated from disclosed token counts, not measured on this project’s own hardware/stack — the same preflight discipline (SPEC 0022’s GO-rule pattern) should apply before spending against them.

6 Recommendation

This is the first path in this analysis line that plausibly meets the \$20-30K target on published, reproducible grounds. Before committing:

1. Pick a teacher (Qwen2.5-7B is the cleanest license fit) and confirm the compression ratio against the 3B target first — the best-tested case.
2. Run a short preflight (pruning + a few hundred million tokens of distillation) to measure real throughput and early quality signal before committing the full 94B-token budget.
3. Scope the Sanskrit fold-in stage as a separate, cheap, clearly-gated addition — don’t let it block or complicate the primary pruning-and-distillation validation.
4. Treat the 1B target’s cost figure as optimistic given the untested 8x compression ratio; the 3B target carries the least extrapolation risk of the three sizes modeled.